and Google—regarded the rise of the cloud as a new age of computing. But in concept, at least, it was something of a regression to the old mainframe architecture of computing's earliest history, where many users all depended upon a single powerful central core that could only be maintained by an elite cadre of professionals. The world had abandoned this "impersonal" mainframe model only a generation before, once businesses like Dell developed "personal" computers cheap enough, and simple enough, to appeal to mortals. The renaissance that followed produced desktops, laptops, tablets, and smartphones—all devices that allowed people the freedom to make an immense amount of creative work. The only issue was—how to store it?

This was the genesis of "cloud computing." Now it didn't really matter what kind of personal computer you had, because the real computers that you relied upon were warehoused in the enormous data centers that the cloud companies built throughout the world. These were, in a sense, the new mainframes, row after row of racked, identical servers linked together in such a way that each individual machine acted together within a collective computing system. The loss of a single server or even of an entire data center no longer mattered, because they were mere droplets in the larger, global cloud.

From the standpoint of a regular user, a cloud is just a storage mechanism that ensures that your data is being processed or stored not on your personal device, but on a range of different servers, which can ultimately be owned and operated by different companies. The result is that your data is no longer truly yours. It's controlled by companies, which can use it for virtually any purpose.

Read your terms of service agreements for cloud storage, which get longer and longer by the year—current ones are over six thousand words, twice the average length of one of these book chapters. When we choose to store our data online, we're often ceding our claim to it. Companies can decide what type of data they will hold for us, and can willfully delete any data they object to. Unless we've kept a separate copy on our own machines or drives, this data will be lost to us forever. If any of our data is found to be particularly objectionable or otherwise in violation of the terms of service, the companies can unilaterally delete our accounts, deny us our own data, and yet retain a copy for their own records, which they can turn over to the authorities without our knowledge or consent. Ultimately, the privacy of our data depends on the ownership of our data. There is no property less protected, and yet no property more private.

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